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Interplay of rationality and morality in using ChatGPT for academic misconduct

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ABSTRACT

ChatGPT's capability to generate original content on any topic represents a concern for academic institutions, since students can plagiarise the AI-created content with a low probability of being caught. While prior research on academic misconduct emphasized the combined effects of perceived costs and benefits in explaining students' decisions to cheat, the low detectability of ChatGPT-based plagiarism significantly reduces the cost of being caught, making rational choice theory less powerful in explaining academic misconduct in the AI era. Considering the effects of both rational decision-making and moral considerations, this study shows that individuals experiencing moral disengagement are more likely to engage in ChatGPT-based plagiarism. Other significant predictors include perceived benefits, punishment severity, and informal sanctions. Furthermore, moral disengagement enhances the effects of formal sanctions on ChatGPT-based plagiarism. The results have implications for professors, academic administrators, and plagiarism detection software designers to enforce academic integrity in the ChatGPT era.

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1. Introduction

The artificial intelligence (AI) technology comprises machines, computers, and other artifacts that exhibit various aspects of human intelligence, such as information processing, learning, and decision-making (Chen, Chen, and Lin 2020). As a subset of this technology, generative AI (GAI) can use large language models to produce sophisticated and original content after training on massive datasets (Stokel-Walker and Van Noorden 2023). With written prompts, GAI tools are capable of creating text, images, audio, video, 3D models, and synthetic data (Baïdoo-Anu and Ansah 2023). These functional capabilities can significantly disrupt educational processes and lead to both beneficial and harmful consequences (North 2023). ChatGPT, a prominent tool in the realm of generative AI, can create essays on various topics, write programming codes, draft emails and memos, and plan events (Roose 2023). Its potential benefits for student involvement, collaboration, and distance learning are well documented (Cotton, Cotton, and Shipway 2023). In addition to the demonstrated good performance on standard tests, such as the U.S. Medical Licensing Exam, bar exam, GRE, and psychology tests (Kung et al. 2023; Uludağ, Jiao, and Zhao 2023; Weiss 2023), ChatGPT attracts students and educators by its productivity and efficiency (Malik et al. 2023; Mogavi et al. 2023; Tian et al. 2023).

A global survey from five countries indicated that 61% of the students believe ChatGPT should be used to assist with assignments (Ibrahim et al. 2023).

While its potential benefits are undeniable, there are concerns about its potential misuse (Chan 2023; Mogavi et al. 2023; Petricini, Wu, and Zipf 2023; Smolansky et al. 2023). Educators are concerned that ChatGPT has made it easier and more tempting for students to engage in academic dishonesty, hindering the development of fundamental skills and endangering conventional educational assessment methods (Herman 2022; Mohammadkarimi 2023). The same global survey indicated that 72% of professors believe ChatGPT's use in student assignments should be considered plagiarism (Ibrahim et al. 2023). Hence, while no one can deny that ChatGPT and other AI instruments will continue to offer benefits in educational settings, legitimate concerns exist regarding the extent, to which it could be used in deviant manners among students, and how it affects educators' abilities to ensure learning outcomes (Keegin 2023).

Existing research on students' perceptions of ChatGPT and academic integrity shows that while it is capable of enhancing learning practices (Malik et al. 2023), its use may result in academic integrity violations (Petricini, Wu, and Zipf 2023). Students clearly disapproved of explicit use of AI-generated content, but

had more ambivalent attitudes toward subtler uses of AI, such as paraphrasing the output and utilising shorter amounts of text (Chan 2023). Students with certain prominent personality traits (low conscientiousness, Machiavellianism, extraversion, etc.) are more likely to engage in ChatGPT-based plagiarism (Greitemeyer and Kastenmüller 2023). Furthermore, students with past misconduct behaviours are more likely to consider using AI tools in completing assignments, and less likely to perceive their use as plagiarism (Tindle et al. 2023). These findings imply that moral considerations may be important in ChatGPT's misuse in educational settings. However, morality effects on ChatGPT-based plagiarism have hardly been examined in extant research. Additionally, while most existing studies provide insights into students' perception of ChatGPT use, many lack a strong theoretical foundation (Chan 2023; Petricini, Wu, and Zipf 2023; Tindle et al. 2023). Furthermore, despite students recognising the unethical nature of copying AI-generated content, they struggle with the less clear-cut threshold of AI use in academic essay writing (Chan 2023).

Among various assignment types (e.g. standardised exams, presentations, role plays, written essays, case studies, term projects, etc.), written essays require higher-order cognitive processing levels since they are more likely to require deep learning strategies such as analysis, interpretation, application, and comprehension (Scouller 1998). Outsourcing these assignments to ChatGPT, capable of producing contextually coherent and nearly identical to human writing text based on a set of parameters and prompts (Cotton, Cotton, and Shipway 2023; Mogavi et al. 2023; Rudolph, Tan, and Tan 2023a), can eliminate the need for complex cognitive processing and higher-order thinking, limiting educational value of these assignments. Furthermore, detecting plagiarism in students' writing is challenging, since students who use AI to write assignments can paraphrase them using other tools (Mohammadkarimi 2023). Therefore, among all assessment types, the focus of this study is written essays as the assignment most impacted by ChatGPT (Smolansky et al. 2023).

The current study offers three contributions. First, it is among the first studies examining students' morality in the ChatGPT context. As implied by Tindle et al. (2023), a student's moral standard could play a significant role in the future usage of AI tools. Specifically, we investigate the role of moral disengagement in explaining students' attitudes and intentions. Second, the study proposes and tests an integrative theoretical framework, combining moral disengagement and rational choice theories. While ChatGPT can be considered yet another high-tech plagiarism tool, its ability

to produce original content significantly hinders its detectability, making the external punishment-related costs negligible compared to the student-obtained benefits of achieving higher grades. Therefore, combining rationality and morality mechanisms to investigate how they interact in propelling students to, or preventing them from, using ChatGPT as a plagiarism tool, could offer a valuable contribution. Moreover, the study extends the theory of moral disengagement by identifying it not only as an important predictor for ChatGPT-based plagiarism, but also as a moderator between formal sanctions and future ChatGPT-based plagiarism. Third, in addition to examining the students' intention to use ChatGPT for written assignments, we measured the ChatGPT plagiarism threshold – the perceived acceptable percentage of content generated by ChatGPT to be used in plagiarism. As indicated by Chan (2023), students have an ambivalent perception of AI-related plagiarism. This measure could accurately reflect their comfort level with incorporating AI-generated content into their written assignments.

2. Relevant literature and theoretical background

2.1. Plagiarism and ChatGPT

Various academic dishonesty (misconduct) forms, broadly defined as 'students giving or receiving unauthorized assistance in an academic exercise or receiving credit for work which is not their own' (Kibler 1993, 5), have been encountered since ancient civilisations. This study specifically focuses on plagiarism, since this academic misconduct type is more relevant in a ChatGPT context, compared to other academic dishonesty manifestations, such as lying to professors and staff, taking credit for other students' work, fabricating official excuses, unauthorised collaboration, forging data, etc. (Ferro and Martins 2016).

Broadly defined as the act of appropriating ideas and words from others and passing them as one's own (Hafsa 2021; Park 2003), and misrepresenting one's own effort in creating a written work (Perkins, Basar Gezgini, and Gordon 2019), plagiarism is a serious form of academic misconduct, occurring at both the undergraduate and graduate levels (Decoo 2002). A 2019 survey showed that about 63.5% of the students reported engaging in at least one plagiaristic act (Curtis and Tremayne 2021). Plagiarism has been mainly investigated using two theoretical lenses: rational choice theory, which manifested as a cost–benefit calculation as the driver of plagiarism (Freiburger et al. 2017; Ogilvie and Stewart 2010), and the theory of moral

development, whereby internalised social norms determined students' likelihood to plagiarise (Levine, Kohlberg, and Hewer 1985). Digital technologies and the internet's incorporation into educational practices have significantly simplified and exacerbated plagiarism (Ercegovic and Richardson 2004). The free availability of digital resources makes it much easier than ever for students to copy and paste information from other sources into their own work, amplifying the problem of 'digital plagiarism' and reducing the cost component of the rational theory approach (Parks et al. 2018). To ensure academic integrity, university faculty extensively utilise plagiarism detection software, such as Turnitin, Copyleaks, Unicheck, or Grammarly (Sun and Hu 2022). While detection software use has been shown to reduce plagiarism levels among students (Batane 2010), it may ineffectively identify ChatGPT-created content. Generative AI, such as ChatGPT, uses a large corpus of texts from the internet to create original-sounding content that does not match any single document from the corpus (Bran et al. 2023). Additionally, a user can 'regenerate response' to produce multiple unique AI responses and choose among the output.

While some faculty find that the text generated by ChatGPT lacks logic, accuracy, and originality and contains fabricated references (Dwivedi et al. 2023), it is generally agreed that essays generated by ChatGPT are well-written, with fitting examples and persuasive arguments (Huang 2023; Vázquez-Cano et al. 2023). Given its frictionless 'regenerate response' feature, ChatGPT can write different essays using the same prompt. This capability has led students to present AI-generated text as their own work for university assessments (Huang 2023; Nolan 2023). Although Turnitin has launched a new feature to detect AI use, the accuracy rate was initially relatively low (Fowler 2023), affecting its usage as a deterrent. Other AI-detection platforms vary in accuracy, making detection a challenge (Demers 2023). Existing comparative studies indicate that most AI-detection platforms are generally inaccurate, with a bias toward attributing AI-generated content to humans (false negative) (Weber-Wulff et al. 2023). Other studies propose tiered plagiarism detection platform systems based on accuracy. For example, Walters (2023) tested 16 different AI-detection platforms and concluded that Copyleaks, Turnitin, and Originality.ai had very high accuracy levels, while Writer, Sapling, and Content-Dector.ai had low accuracy levels at the time of testing. As AI software becomes more sophisticated, its detection will become even more challenging, increasing AI-generative platform attractiveness to those with ill intent (Sadasivan et al. 2023).

Existing empirical research on digital (or cyber) plagiarism supports the cost-benefit calculation of the rational choice theory, emphasizing the combined effects of perceived costs (punishment certainty and severity) and benefits (e.g. better grades) in explaining student plagiarism decisions. Students plagiarise digital content for efficiency gains and to get higher grades (Park 2003; Power 2009), to compete with (or emulate) peers (Ramzan et al. 2012; Šprajc et al. 2017), and are deterred from plagiarising by the perceived risks of detection and formal punishment (Davis and Ludvigson 1995; Nwosu and Chukwuere 2020). Extant papers also separately examine roles of ethics and morality (Mulholland 2020). For instance, a recent meta-analysis of nursing student plagiarism perceptions showed that most students (88%) perceived plagiarism as an unethical behaviour (Fadlalmola et al. 2022). Their perceptions differed by plagiarism type. Namely, plagiarising all or most of the work was rated very unethical while copying a few sentences was rated the least unethical by students in the United States and Canada (McCabe 2009).

The low detectability of ChatGPT-based plagiarism reduces the cost of being caught and renders rational choice theory less powerful in explaining academic misconduct in the AI era. Therefore, it is logical to explore the combined effects of costs, benefits, and ethical/moral predispositions in driving students' intention to use ChatGPT for plagiarism.

2.2. Moral disengagement theory

Moral disengagement (MD) refers to a cognitive process of decoupling a person's internal moral standards from unethical or criminal behaviour without experiencing distress (Moore 2015). According to the MD Theory (Bandura 1986), MD combines several cognitive mechanisms (e.g. attribution of blame to another, advantageous comparison and euphemistic labelling to trivialise the infringement, consequence distortion, responsibility displacement, etc.) that allow individuals to perceive deviant behaviours as unrelated to their moral standards, rationalising their actions and alleviating guilt. MD theory has been applied and supported in multiple contexts, including illegal downloading (Olivero et al. 2019), criminal behaviour (Cardwell et al. 2015), ethical decision-making (Wu, Hu, and Li 2023), and academic cheating (Farnese et al. 2011). MD is theorised to operate both as a process, and as a disposition (moderator) in individuals' responses to morally problematic opportunities (Moore 2015). MD was found to significantly predict academic misconduct intention (Farnese et al. 2011; Fida et al. 2018). Earlier research also finds that students with weak moral

views on cheating appear to more readily engage in various academically dishonest behaviours (Whitley 1998). Surprisingly, MD's role has been insufficiently explored in a digital academic misconduct context despite abundant opportunities for students to cheat and plagiarise from digital sources in any place and at any time, with blurred lines between original and borrowed content. Al-Nuaimi and Uzun (2023) note that the COVID-19 pandemic shifted educational emphasis to online learning and provided greater plagiarism opportunities. After finding an insignificant effect of moral obligation on plagiarism intention during COVID-19 in the online learning format, the authors suggest that moral disengagement may mediate the moral judgment – dishonest academic actions relationship. In accordance with earlier findings of MD's positive direct role in transgressive behaviours, including academic dishonesty, we hypothesise that:

H1a: Moral disengagement is positively related to the intention to use ChatGPT for plagiarism.

Traditional plagiarism detection software (e.g. Turnitin) matches submitted written assignments to available textual sources in various databases. It produces a percentage-based similarity index (or an originality report) showing the amount of content copied from already existing documents (Balbay and Kilis 2019). However, ChatGPT can produce unique and original content, with very low similarities to existing bodies of text (Sadasivan et al. 2023). Since educators generally accept a certain degree of similarity in student submissions due to direct citations or repeated phrases in the same submission, students may feel that using lower amounts of ChatGPT-generated output can reduce the likelihood of plagiarism detection and the consequent punishment. In addition, people often behave dishonestly for personal gain but still desire to maintain their honest self-image (Mazar, Amir, and Ariely 2008). According to the theory of self-concept maintenance (Mazar, Amir, and Ariely 2008), people who are more concerned with maintaining their ethical self-concept would try to lower the degree of their transgression. People's sense of morality is related to the cheating amount that they are comfortable with (Ariely and Jones 2012). Therefore, it is logical to suggest that individuals who are morally disengaged, and thus, do not attribute ChatGPT plagiarism to their moral self-perception, would tolerate a higher percentage of ChatGPT-generated output in their essays than those who do not separate their academic misconduct from their moral self-perceptions. Hence, we use the term 'ChatGPT plagiarism threshold' to indicate the perceived acceptable

percentage of content generated by ChatGPT to be used in plagiarism. Therefore, we propose:

H1b: Moral disengagement is positively related to the ChatGPT plagiarism threshold.

2.3. Rational choice theory: perceived benefits

Attributed to Adam Smith's eighteenth-century work, rational choice theory proposes that calculated self-interest is the basis of individual decision-making, and that people juxtapose perceived benefits against perceived costs in making behavioural choices (Scott 2000). Specifically, deviant non-pathological behaviours are a function of individual subjective calculations of the perceived consequences' costs and benefits (Clarke and Felson 1993). Regarding academic misconduct, plagiarism's benefits could include higher grade expectations, producing better quality assignments, and saving time (Ogilvie and Stewart 2010). Academic misconduct costs can be perceived as punishments, including academic sanctions (e.g. low or failing grades or expulsions) (Covey, Saladin, and Killen 1989) and the internally imposed anticipated feelings of guilt, shame, or embarrassment (Cochran et al. 1999). Validation exists for rational choice theory in an academic misconduct context, with Williams and Hosek (2003) indicating that students are rational decision-makers and decide to cheat when the perceived benefits outweigh the perceived costs of being caught.

Obtaining plagiarism's benefits (higher grades and time-saving) is facilitated by modern digital technologies that offer instant 'cut and paste' capabilities, abilities to search the required content in vast 'cloud' data depositories, and the possibility to ask anonymous questions online. This 'virtualization' (removing physical barriers to locating and illicitly copying of material), as well as the introduction of artificially intelligent GPT technologies, allows students to avoid such traditional plagiarism barriers as requirements to submit paper drafts, turn in research materials, or more specific assignment design (Kauffman and Young 2015). Conversely, ChatGPT's plagiarism benefits are expanded by incorporating the output's uniqueness and originality, quick access to the world's digital archives, and the tool's limitless availability, providing additional incentives for student plagiarism (Sok and Heng 2023). ChatGPT's constant availability, proficiency, and near-undetectability can lead to student overreliance on the tool by using it to produce entire assignments, especially for last-minute submissions, increasing perceived benefits. Therefore, we propose:

H2a: Perceived benefits are positively related to the intention to use ChatGPT for plagiarism.

H2b: Perceived benefits are positively related to the ChatGPT plagiarism threshold.

2.4. Rational choice theory: formal sanctions

General deterrence theory is often used to explain deviance abstention. Assuming individuals are rational beings, the theory poses that if the expected cost is greater than the expected gain, people will avoid deviant behaviours. The theory's core cost components are punishment severity and certainty (Becker 1968), usually called formal sanctions. Any increases in perceived punishment severity and/or certainty will increase the cost of deviant behaviour, as shown in academic dishonesty literature (Cochran et al. 1999; Freiburger et al. 2017).

Perceived severity of formal sanctions refers to the perceived magnitude of the externally imposed penalty if caught. In our context, it is the students' perception of the punishment severity that they receive if they are caught submitting ChatGPT-generated essays. To deter plagiarism, universities enforce policies to deter these activities with formal punishment. A high level of perceived punishment severity tends to increase the perceived cost of plagiarism. Plagiarism cannot be deterred if the penalty is not harsh. Researchers have found a negative relationship between perceived punishment severity and academic misconduct (Teixeira and de Fátima Oliveira Rocha 2010). Therefore, we hypothesise that:

H3a: Perceived severity of formal sanctions is negatively related to the intention to use ChatGPT for plagiarism.

H3b: Perceived severity of formal sanctions is negatively related to the ChatGPT plagiarism threshold.

Perceived certainty of formal sanctions measures the probability of an individual receiving the imposed penalty. When the punishment probability is low, punishment severity will have little or no effect on deterring plagiarism. Nevertheless, high perceived punishment certainty tends to increase the costs of plagiarism. There is evidence that students' perception of punishment certainty reduces the likelihood of academic misconduct (Cochran 2015; Freiburger et al. 2017). When plagiarism cannot be detected, as is plausibly the case with the ChatGPT-generated content, students could perceive a low level of punishment certainty and will feel encouraged to engage in academic misconduct. In a similar vein, students' belief in lower plagiarism detection, if less ChatGPT plagiarised content is used, could prompt reduced ChatGPT content utilisation for plagiarism. Therefore, we hypothesise that:

H4a: Perceived certainty of formal sanctions is negatively related to the intention to use ChatGPT for plagiarism.

H4b: Perceived certainty of formal sanctions negatively relates to the ChatGPT plagiarism threshold.

Despite the negative relationships between formal punishment severity and certainty and academic misconduct (Cochran 2015; Freiburger et al. 2017; Teixeira and de Fátima Oliveira Rocha 2010), a few studies suggest that formal sanctions do not exhibit any significant relationships with academic misconduct (Cochran et al. 1999; Ogilvie and Stewart 2010). Additionally, Nagin and Pogarsky (2003) reported lower cheating prevalence on a quiz when detection was more certain but not when the penalty was more severe. To reconcile these views, we propose that moral disengagement may moderate the relationship between formal sanctions and academic misconduct. While inherently moral individuals would avoid dishonesty, regardless of the magnitude or probability of formal punishments (costs), a more amoral person will cheat as long as potential gains exceed the potential costs (Williams and Hosek 2003). However, most individuals appear to fall between these extremes. Relying on the theory of self-concept (Mazar, Amir, and Ariely 2008), some authors suggest activating one's positive self-concept will reduce dishonesty and cheating (González-Arango, López-Ardila, and Corredor 2020; Hertwig and Mazar 2022), which would disbalance the cost-benefit analysis when considering deviant behaviours (Shu, Gino, and Bazerman 2011). Analogously, we propose that moral disengagement would amplify the perceived punishment's role (both the severity and the certainty components) so that:

H5a: The negative relationship between the perceived severity of formal sanctions and the intention to use ChatGPT for plagiarism will be strengthened as moral disengagement increases.

H5b: The negative relationship between the perceived severity of formal sanctions and the ChatGPT plagiarism threshold will be strengthened as moral disengagement increases.

H6a: The negative relationship between the perceived certainty of formal sanctions and the intention to use ChatGPT for plagiarism will be strengthened as moral disengagement increases.

H6b: The negative relationship between the perceived certainty of formal sanctions and the ChatGPT plagiarism threshold for plagiarism will be strengthened as moral disengagement increases.

2.5. Informal sanctions

Existing criminology research suggests that deviance deterrence is not limited to formal or legal factors. For

instance, Grasmick and Bursik (1990) identified attachment to socially significant others and internalised moral norms as internal sources of punishment that affect decision-making as strongly as formal punishment. Correspondingly, education research shows that self-imposed informal sanctions (such as shame, guilt, embarrassment, or social exclusion) play a strong role in deterring academic cheating (Tibbetts 1998). Furthermore, Cochran et al. (1999) found that while neither severity, nor certainty of formal sanctions for engaging in academic misconduct significantly impacted academic misconduct, the likelihood of experiencing shame significantly reduced student plagiarism. These results emphasize the importance of internal, self-imposed sanctions in deterring academic transgressions and affirm their role as psychological costs of academic dishonesty (Nagy and Groves 2021). Detection difficulties regarding ChatGPT-based plagiarism can render formal sanctions less effective, amplifying the deterrent role of internal psychological costs of embarrassment, shame, or guilt experiences. Along with the aversive moral emotions' role in ChatGPT-based plagiarism deterrence, the 'fudge factor' perspective (Mazar, Amir, and Ariely 2008) suggests that individuals will cheat only insofar as they are able to maintain a positive self-image. Thus, it is suggested that the internal need to maintain a positive self-image may reduce the amount of content plagiarised with ChatGPT's help. Therefore:

H7a: Perceived informal sanctions are negatively related to the intention to use ChatGPT.

H7b: Perceived informal sanctions are negatively related to the ChatGPT plagiarism threshold.

Figure 1 depicts the proposed research model.

3. Methodology

3.1. Study design

A survey was administered via Qualtrics to a Prolific panel of US respondents to test the hypotheses.¹ Prolific, an established platform for online subject recruitment, is chosen because the platform is specifically designed for scientific studies and has clear guidelines on the rights, obligations and compensation of respondents (Palan and Schitter 2018). Compared to other online respondent platforms maintained by commercial survey firms and in-person studies with undergraduate students, Prolific respondents yield higher data quality (Douglas, Ewell, and Brauer 2023; Peer et al. 2022). In particular, Prolific elicits respondent characteristics independently of specific studies so researchers can pre-screen respondents based on these characteristics (Palan and Schitter 2018). To enhance data quality, we implemented guidelines in accordance with recommendations from seminal work regarding online panels (Goodman and Paolacci 2017; Matherly 2019). First, respondents were restricted to English-speaking US-based respondents. Second, the respondent approval rating was set to 99+%. Additionally, security settings restricted responses to one response per IP address. Given the study's context, a qualifier was included regarding their current student status, and a question was included to remove respondents without prior ChatGPT awareness. Finally, an attention check was incorporated into the survey to eliminate inattentive respondents.

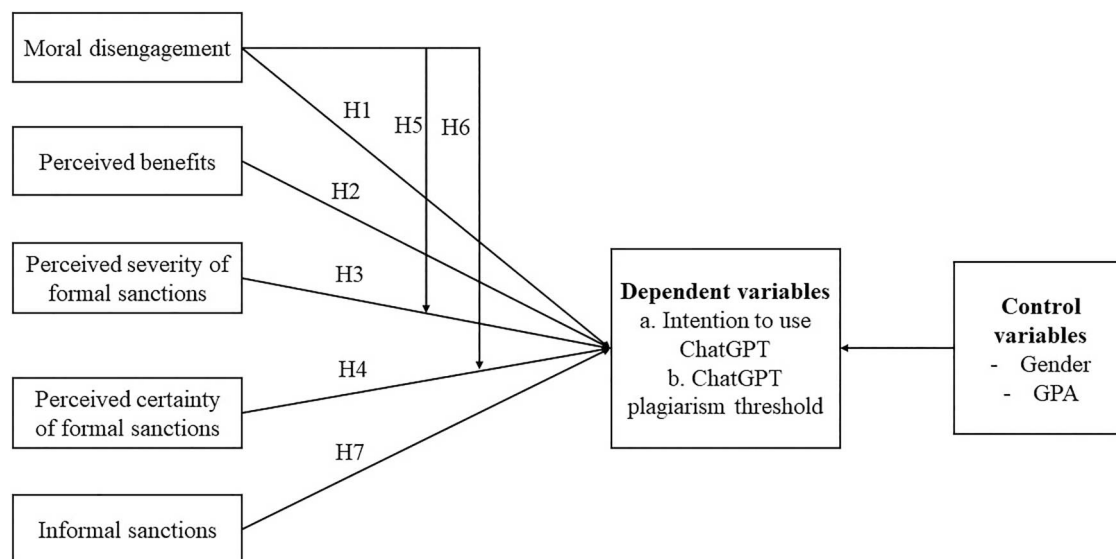


Figure 1. Research model.

3.2. Sample characteristics

Since small effects are usually observed in the tests of moderation, a power analysis demonstrated that a sample size of 485 offered sufficient power (> 0.8) to detect a small effect size with an alpha level of $p = 0.05$ (Faul et al. 2009). The Prolific Panel data collection resulted in 504 respondents. Six respondents didn't have prior ChatGPT awareness, four respondents potentially were associated with the same IP address, and three respondents were eliminated due to Prolific ID issues. Therefore, thirteen responses were eliminated, and all remaining respondents passed the attention check, asking them to select from a pre-specified option in the instructions. Of the 491 remaining respondents, 55% were male, 41% were female, and 4% non-binary/prefer not to say. They were 53% Caucasian, 21% Black/African American, 10% Asian, 10% Hispanic, and 6% Multi-racial/Native American or Pacific Islander. Regarding age, 28% of respondents are younger than 23, 24% between the ages of 24 and 29, 23% between the ages of 30 and 39, and 25% older than 39. In terms of institution, 58% were attending a public university, 20% a private university, 13% a community college, 5% an online university, and 4% a technical college/other institution. Regarding majors, they were 20% business, 15% computer and information science, 12% health-related fields, 12% social sciences, and 9% engineering, with the remaining 32% spread across more than 14 different majors or undecided. Respondents were asked about usage behaviour, and 87% indicated ChatGPT usage. In ChatGPT usage terms, 33% used it for brainstorming/idea generation, 19% used it to answer multiple choice questions, 19% detected errors with their own solutions, 18% solved short essay questions, 15% submitted ChatGPT-created materials with edits, 10% used it to debug codes, and 4% submitted ChatGPT created written materials without edits.

3.3. Primary measures

Two measures were used to represent the dependent variables of intended ChatGPT use. The first variable, ChatGPT use intention, consists of three items (e.g. I intend to use ChatGPT for written assignments in the future) measured on a 1–7 Likert scale ($\alpha = .97$). The second dependent variable, ChatGPT plagiarism threshold, asked respondents to indicate an acceptable percentage (0–100%) of ChatGPT-generated content to be used in a submitted written assignment.

Five independent variables were measured for the study. First, Shu et al.'s (2011) six-item scale captured moral disengagement on a 1–7 Likert scale ($\alpha = .84$).

Second, three items were developed to measure perceived ChatGPT benefits on a 1–7 Likert scale ($\alpha = .84$). The third measure used a 1–7 itemised rating scale and consisted of two items focusing on perceived punishment severity for ChatGPT usage ($\alpha = .91$). The fourth independent variable focused on punishment certainty, also measured using two items on a 1–7 itemised-rating scale ($\alpha = .91$). Finally, Bossler's (2021) three-item informal sanctions measure was adapted using a 1–7 Likert scale, capturing perceived informal sanctions if caught using ChatGPT ($\alpha = .89$). All scale items are in the Appendix.

Beyond the independent and dependent variables, two validated marker variables were included for common method variance analysis (Simmering et al. 2015). The first marker variable captures one's attitude towards blue and was developed by Miller and Chiodo (2008) as an effective marker variable. The measure uses a three-item Likert scale ($\alpha = .82$). The next marker variable (Hansen 2012), internet usage ($\alpha = .88$), captures general internet search behaviour using three items on a 1–5 itemised rating scale.

3.4. Common method variance

To account for common method variance (CMV), the following steps and assessments were implemented. According to Podsakoff, MacKenzie, and Podsakoff (2012), the key to mitigating CMV is to ensure that respondents are capable of providing accurate responses. Therefore, the following practices were implemented. First, instructions were clear and concise, and the scales used were straightforward in design to not require high cognitive effort levels. Composite scales were six items or less. CMV is most prevalent when respondents must exert considerable effort processing question information and retrieving information from memory and, thus, resort to satisficing responses. The survey was pre-tested with college students prior to launching the main study to identify any issues, and it was estimated that it would generally take no more than six minutes to respond to the survey, increasing the likelihood that respondents spent more time per question (SurveyMonkey 2012). The median time for the main study was 311 s (5 min, 11 s). Additionally, the sample was restricted to current students as they would likely have the awareness and motivation to provide input on ChatGPT usage. We eliminated respondents with no ChatGPT awareness and inserted a short ChatGPT refresher video into the survey for respondents who indicated awareness (13% of the sample), but no prior usage.

To examine CMV protocol effectiveness, we used the recommended marker technique (Podsakoff,

MacKenzie, and Podsakoff 2012; Simmering et al. 2015) since Harman's One-Factor test is deficient in detecting CMV (Baumgartner, Weijters, and Pieters 2021). The blue attitude and internet usage marker variables chosen were deemed to be theoretically unrelated to the current variables and, thus, should not have any significant associations. The Pearson correlation coefficient was used to examine the association strength among the variables, including the two marker variables. For the blue attitude marker, no significant correlations occurred (ps : .28 to .99). For the internet usage marker, all correlations were non-significant (ps : .06 to .72) except for punishment severity ($p < .05$). CMV-adjusted correlation values were calculated according to Malhotra, Schaller, and Patil (2017), and all correlations between substantive variables remained significant ($ps < .001$).

Subsequently, regression analyses were run with and without the marker variables. For the intention to use ChatGPT analysis, all significant relationships were unaffected by the inclusion of the marker variables. R^2 also was unaffected by their inclusion (.476 vs. .474). A similar pattern was observed for analysis using the ChatGPT plagiarism threshold dependent variable (R^2 : .431 vs. .436). The analyses from both variables suggest that CMV minimally affected the results.

4. Results

SPSS 29 was used for all analyses, and all continuous independent variables were mean-centered for the analyses. We first examined if there were significant differences between the two dependent variables across the demographic groups. There were no significant differences in respondents by university type ($p > .45$). However, there were significant differences in respondents by major. A closer look revealed that students majoring in computer sciences, engineering, business, and health-related fields had a higher intention to use ChatGPT and a higher ChatGPT plagiarism threshold than students who majored in social sciences, education and other majors.

Caucasians had lower intention to use ChatGPT than others ($p < 0.01$) while males had significantly higher ChatGPT intention levels than other genders ($p < 0.01$). Regarding plagiarism threshold, Caucasians reported a significantly lower plagiarism threshold than other ethnicities ($p < 0.001$). Males reported a higher ChatGPT plagiarism threshold level than other genders ($p < 0.05$).

Table 1 shows the descriptive statistics for all variables. Linear regression was used to analyse the hypothesised six independent variables' effects and the two interaction effects on the intention to use ChatGPT. Gender identity (male vs. other) and GPA were included as variables to control for their potential effects (Lee, Kuncel, and Gau 2020). Ethnicity is also included as a control variable since it was found to be significantly related to the two dependent variables.

First, Models 1 (main effects) and 2 (interactions) demonstrated the following independent variable effects on intention to use ChatGPT. Moral disengagement (H1a: $\beta = .26$; $p < .001$) and ChatGPT benefits (H2a: $\beta = .43$; $p < .001$) were positively related, while punishment severity (H3a: $\beta = -.19$; $p < .001$), punishment certainty (H4a: $\beta = -.11$; $p < .05$) and informal sanctions (H7a: $\beta = -.24$; $p < .001$) were negatively related to intention to use ChatGPT. Further in Model 2, a significant interaction occurred between moral disengagement*punishment certainty (H6a: $\beta = -.07$; $p < .05$), but not for moral disengagement*punishment severity (H5a: $p > .48$). Regarding control variables, no significant effects were observed ($ps > .13$) in the two models (See Table 2).

To further elucidate the moral disengagement*punishment severity interaction effects on ChatGPT plagiarism threshold, the sample was grouped using cluster analysis into high ($M: 4.18$) and low ($M: 2.13$) moral disengagement groups. As shown in Figure 2, for high moral disengagement respondents, punishment certainty perceptions were more influential as a deterrent for future ChatGPT usage, approaching the future usage ratings of the low moral disengagement group,

Table 1. Descriptive statistics and correlations.

	1	2	3	4	5	6	7	8	9
GPT Future Usage	0.97								
GPT Essay %	.63***	NA							
Moral Disengagement	.42***	.42***	0.84						
Perceived Benefits	.57***	.47***	.32***	0.84					
Punishment Severity	-.40***	-.47***	-.21***	-.28***	0.91				
Punishment Certainty	-.43***	-.39***	-.34***	-.36***	.46***	0.91			
Informal Sanctions	-.53***	-.47***	-.36***	-.42***	.43***	.46***	0.89		
Blue Marker	0.01	0.02	0.05	0.05	0.00	0.03	0.00	0.82	
Internet Use Marker	-0.02	-0.08	0.02	0.04	.11*	-0.09	-0.05	.10*	0.88
Mean	3.06	20.85	3.18	4.33	5.38	4.50	4.72	5.17	4.43
SD	2.01	21.62	1.24	1.62	1.36	1.60	1.98	1.29	0.64

Note: ***, **, * represent significant correlations at the level of .001, .01, and .05, respectively. Cronbach α s are in italics.

Table 2. Results of multiple regression analyses.

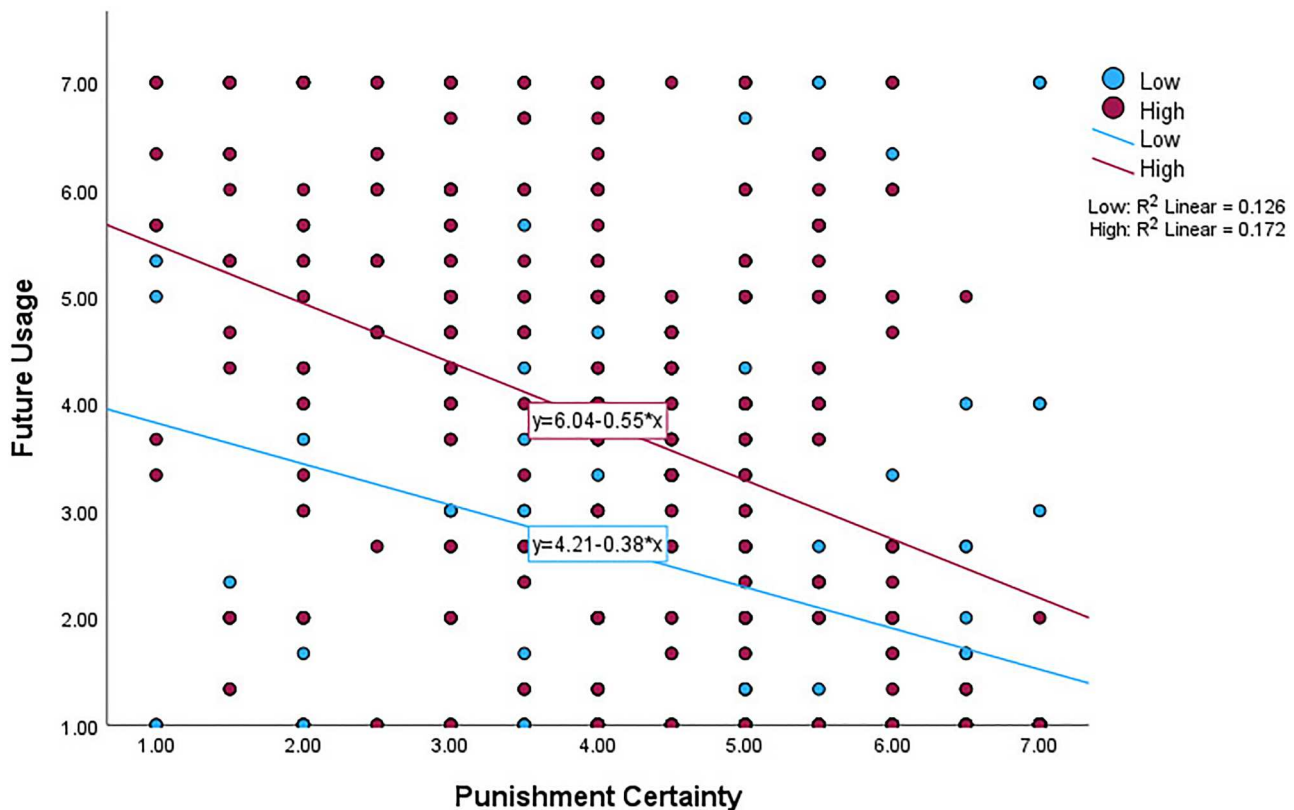
Independent variable	Future GPT usage Model (1)	Future GPT usage Model (2)	Plagiarism threshold Model (3)	Plagiarism threshold Model (4)
Moral Disengagement	0.26***	0.26***	3.68***	3.66***
ChatGPT Benefits	0.43***	0.43***	3.21***	3.19***
Punishment Severity	-0.19***	-0.20***	-4.34***	-4.05***
Punishment Certainty	-0.11*	-0.11*	-.40	-0.62
Informal Sanctions	-0.24***	-0.22***	-1.72***	-1.82***
GPA	0.05	0.05	0.75	0.80*
Male	0.04	0.06	-0.48	-0.53
Caucasian	-0.14	-0.16	-3.44*	-3.32*
Moral Disengagement * Punishment Severity		0.03		-1.55**
Moral Disengagement * Punishment Certainty		-0.07*		-.12
N	491	491	491	491
Adj. R ²	0.48	0.48	0.42	0.43
F Value	56.23	45.59	45.63	38.16

Note: ***, **, * represent significant correlations at the level of .001, .01, and .05, respectively.

as punishment certainty increased. In sum, hypotheses H1a, H2a, H3a, H4a and H6a were supported, while no support was found for H5a.

Next, the hypothesised effects on the ChatGPT plagiarism threshold were examined in Models 3 (main effects) and 4 (interactions). Model 3 regression results revealed positive effects for moral disengagement (H1b: $\beta = 3.68$; $p < .001$) and perceived benefits (H2b: $\beta = 3.21$; $p < .001$), while negative effects were observed for punishment severity (H3b: $\beta = -4.34$; $p < .001$) and informal sanctions (H7b: $\beta = -1.72$; $p < .001$). For the interactions, Model 4 indicated a significant interaction for moral disengagement* punishment severity (H5b: $\beta = -1.55$; $p < .01$), but not moral disengagement*punishment certainty (H6b: $p > .77$). Regarding control variables, significant effects were observed for Caucasian ($p < .03$) and GPA ($p < .04$) in the two models.

To further examine the moral disengagement*punishment severity interaction effects on the ChatGPT plagiarism threshold, the same two moral disengagement groups from Figure 2 were used. As conveyed in Figure 3, the effects of perceived punishment severity on the ChatGPT plagiarism threshold are stronger for the high moral disengagement group than for the low moral disengagement group. Hence, the high moral group's responsiveness, as measured through the ChatGPT essay % threshold, to punishment severity approaches the low morality group's, when punishment is perceived as more severe. Overall, the data supported hypotheses H1b, H2b, H3b,

**Figure 2.** Interaction effect between punishment certainty and moral disengagement on ChatGPT usage.

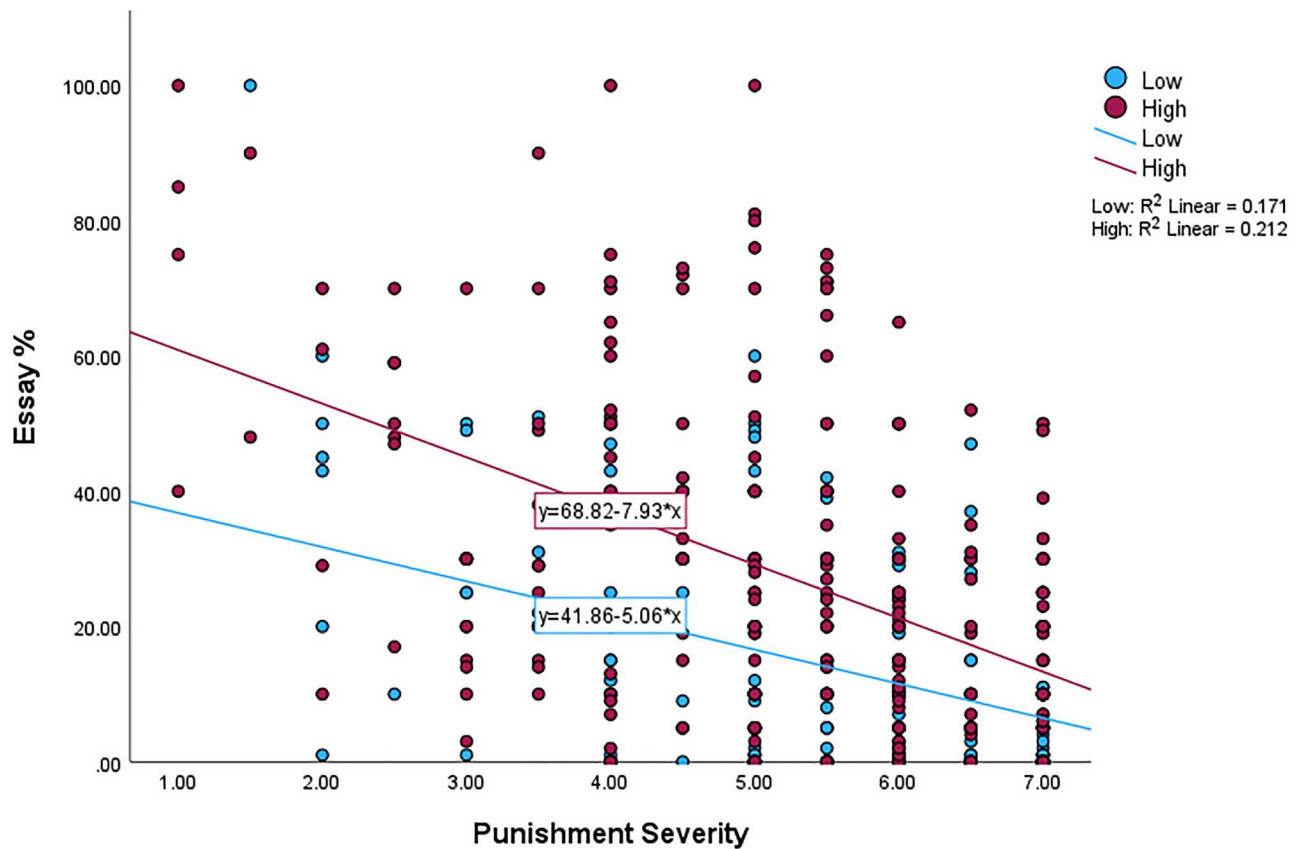


Figure 3. Interaction effect between punishment severity and moral disengagement on ChatGPT plagiarism threshold.

and H5b, while H4b and H6b were unsupported. See Table 3 for the hypotheses testing summary.

5. Discussion

The goal of this research was to examine factors affecting students' intentions of using ChatGPT for plagiarism and the extent (threshold) of the intended

plagiarised content. Our results reveal that perceived benefits (such as passing a class or obtaining a higher grade on the assignment) positively affect students' intention to use ChatGPT for plagiarism and their ChatGPT plagiarism threshold. This finding supports earlier academic misconduct studies in other contexts and provides evidence for rational choice theory in explaining AI-based plagiarism behaviours (Parks et al. 2018; Teixeira and de Fátima Oliveira Rocha 2010). When the perceived benefits of using ChatGPT are high, individuals are more likely to use the software for potential gains.

Further supporting the rational choice theory, our results show that punishment severity perceptions are negatively related to intentions to use ChatGPT and the ChatGPT plagiarism threshold. When individuals perceive that the consequences of using ChatGPT are severe, such as failing a course, academic probation, or suspension from the institution, they are more likely to be deterred from using ChatGPT for plagiarism and have a higher ChatGPT plagiarism threshold. The deterrent effect is based on the rational calculation that the severity of punishment outweighs the potential benefits of using ChatGPT.

We also find a significant effect of punishment certainty on the intention to use ChatGPT. The certainty

Table 3. Test of hypotheses.

Hypotheses	Relationship	Supported
H1a:	Moral Disengagement → Plagiarism intention	Yes
H1b:	Moral Disengagement → Plagiarism threshold	Yes
H2a:	Benefits → Plagiarism intention	Yes
H2b:	Benefits → plagiarism threshold	Yes
H3a:	Punishment Severity → Plagiarism intention	Yes
H3b:	Punishment Severity → Plagiarism threshold	Yes
H4a:	Punishment Certainty → Plagiarism intention	Yes
H4b:	Punishment Certainty → Plagiarism threshold	No
H5a:	Moral Disengagement * Punishment Severity → Plagiarism intention	No
H5b:	Moral Disengagement * Punishment Severity → Plagiarism threshold	Yes
H6a:	Moral Disengagement * Punishment Certainty → Plagiarism intention	Yes
H6b:	Moral Disengagement * Punishment Certainty → Plagiarism threshold	No
H7a:	Informal Sanctions → Plagiarism intention	Yes
H7b:	Informal Sanctions → plagiarism threshold	Yes

of getting caught and punished for plagiarism is a significant deterrent for academic misconduct. When students estimate a higher probability of getting penalised for engaging in plagiarism using ChatGPT, they are less likely to utilise the tool. Although ChatGPT plagiarism is difficult to detect and plagiarism detection tools are imperfect, informing students of the new and developing AI-detection platforms can lower their intentions to plagiarise. Interestingly, we did not find a significant relationship between punishment certainty and the ChatGPT plagiarism threshold. One possible reason is that respondents do not have a clear understanding of the acceptable level of using ChatGPT in writing. In a survey of university policies on generative AI use, 'copying and pasting AI-generated responses' is the most frequently used term (Moorhouse, Yeo, and Wan 2023). This wording could be misleading, making students assume that adjustments and modifications of AI-generated content will go unnoticed by AI detection tools, making punishment certainty irrelevant for the ChatGPT plagiarism threshold.

Compared to the divergence within the formal cost elements, we find that the deterrent effects of informal sanction were consistently observed for both the intention to use ChatGPT and the ChatGPT plagiarism threshold. Informal sanctions in our study (e.g. shame, embarrassment, etc.) are far more influential than deterrence-based formal sanctions. These emotional responses constitute the social and psychological costs of being caught cheating. It is possible that the social stigma associated with AI-based plagiarism and its potential long-lasting effect on a student's reputation and self-esteem can act as a powerful deterrent to academic misconduct.

We find that moral disengagement intensifies both the intention and the extent of plagiarising using ChatGPT, supporting the explanatory role of moral disengagement theory (Farnese et al. 2011). As predicted, moral disengagement moderated the effects of formal punishment on both the intention and the extent (threshold) of ChatGPT plagiarism. Interestingly, however, it did not affect the relationship between external punishment expectations and plagiarism intentions equally. Punishment certainty was more of a deterrent for ChatGPT usage for the cohort with high level of moral disengagement relative to the low-level group. Regarding the ChatGPT content threshold, individuals in the high moral disengagement group were also more likely to see punishment severity as a deterrent for increasing the amount of ChatGPT content in an essay.

Our study also revealed significant relationships between demographic variables and the use of ChatGPT for plagiarism. Particularly, males reported a higher

intention to use ChatGPT for plagiarism and a higher plagiarism threshold than other genders. This is consistent with previous research, which found significantly higher levels of cheating among males (McCabe and Trevino 1997; Whitley, Nelson, and Jones 1999). The effects of gender on the dependent variables, however, became non-significant after the main independent variables were entered. Regarding race, the descriptive statistics revealed that Caucasians have a lower level of intention to use ChatGPT for plagiarism and a lower threshold for ChatGPT plagiarism than non-Caucasians. The effect of race on the ChatGPT use intention (but not on the plagiarism threshold) became insignificant after the main independent variables were entered. Previous research on the relationship between race and academic integrity yielded mixed results. While Martin, Rao, and Sloan (2011) found no difference in levels of plagiarism between Caucasians and Asians, Lanier (2006) reported a significant difference in cheating among races. In particular, among five races, Caucasians ranked 3rd in the likelihood of cheating in online courses and 4th in traditional courses. Given the findings yielded from the study, we propose that more research be done examining the effects of demographic variables on ChatGPT usage.

6. Implications

6.1. Theoretical implications

This research makes the following contributions to the plagiarism scholarship. First, the paper proposes and tests a comprehensive theoretical framework of academic misconduct drivers and mechanisms in the context of an emerging and extremely powerful ChatGPT technology that has the potential to seriously threaten academic integrity. While the effortless use of IT and the Web has been a substantial driver for plagiarism (Jereb et al. 2018), ChatGPT use can significantly accelerate cheating on written essays, making it virtually invincible to plagiarism detection software. Our study answers the call for academic researchers to examine academic cheating potential and mechanisms in the newly emerging technological environment.

Second, the study supports and refines rational choice theory in an AI-based plagiarism context. We find that informal sanctions are more powerful predictors than formal sanctions, supporting previous literature (Cochran et al. 1999; Grasmick and Bursik 1990). The study further delineates the nuanced impacts of punishment severity and certainty. Our findings reveal that punishment severity is significantly related to the intention to use ChatGPT and ChatGPT plagiarism

threshold, while punishment certainty is only significantly related to the intention. Considering the previous mixed results of the differential roles of formal sanctions' components on deviant behaviours, we suspect that the relative effects of formal sanctions' components may be context-dependent. This supposition should be carefully tested in future studies.

Third, our results support the moral disengagement theory and contribute to understanding the extent of moral disengagement not only as a personality-related driver, but also as a boundary condition. We show that the personal predisposition to moral disengagement not only directly affects the intention to plagiarise and the degree of intended plagiarism, but also moderates the relationship between formal sanctions and ChatGPT-based plagiarism. The results suggest that increasing formal sanctions for cheating can act as a more effective deterrence for ChatGPT-based plagiarism among morally disengaged students.

6.2. Practical implications

Our findings also provide implications for educators, administrators, and plagiarism detection software designers as they work to leverage ChatGPT and other generative AI, while also discouraging generative AI misuse. First, since ChatGPT-based plagiarism's low detectability presents a major challenge, education administrators and professors could emphasize punishment severity as a formal deterrent for ChatGPT-based plagiarism when developing class, college, and institutional policies. Additionally, individual assignments may require closer personal monitoring and/or proctoring. While not ideal, these may be necessary measures until more advanced plagiarism detection tools are developed. Second, educators may rethink how to assess student learning. Traditional assessments, such as individual essay-type assignments, may need restructuring. Specifically, students could be required to submit the assignments gradually, by modules, with each following module containing corrections and changes made based on instructor feedback (Rudolph, Tan, and Tan 2023b). Thus, the effort and time requirements for students may serve as a deterrent for some students to use ChatGPT. Educators may also consider innovative assessment types to reduce plagiarism. For example, assessments can be designed to require students to engage in group discussions, presentations and other interactive activities. Further, authentic case studies and practical assessments that involve contextual and real-world problems may reduce reliance on AI tools (Rudolph, Tan, and Tan 2023b). Moreover, allowing students to write about the subjects that truly interest them may foster

an environment where their voices are heard and opinions valued (McMurtrie 2022). This may reduce its usage from a motivational (benefits) standpoint. Third, universities could introduce a ChatGPT ethics course, as evidence exists for Internet ethics curriculum effectiveness as a deterrent, emphasizing the moral costs of AI-based misconduct (Wang, Wang, and Wang 2022). Finally, plagiarism detection developers should create AI-based, more sophisticated tools to help educators and users identify plagiarised content accurately, enhancing the perception of punishment certainty to deter illegitimate ChatGPT usage.

6.3. Limitations and future research

The study's findings should be interpreted while considering the following limitations. First, we used respondents' self-reported data obtained via a cross-sectional survey design. Thus, no causal inferences can be made. Future studies could use other methodologies, such as experiments, to examine the effects of the proposed variables on ChatGPT usage. For example, the effects of more vs less severe formal punishment policies on student plagiarism using ChatGPT can be explored. Specifically, punishment severity and/or punishment certainty can be manipulated to identify optimal policies discouraging ChatGPT-based academic deviance. One manipulation might be to examine whether requiring students to submit assignments gradually affects punishment certainty perceptions. Also, priming moral engagement can be tested experimentally to understand whether reinforcing moral integrity can reduce ChatGPT-based plagiarism incidents. Furthermore, longitudinal studies can be conducted to examine students' perspectives and attitudes toward ChatGPT-based plagiarism over time. Second, a nonrandom judgment sample recruited via Prolific was used. Future studies can utilise a more representative student sample to improve the results' generalizability. Moreover, the study's sample was limited to US students. Therefore, future research can examine student ChatGPT perceptions and behaviours in other countries. Cultural, political, or religious factors may affect perceptions toward ChatGPT-based plagiarism. Third, while the relationships examined in the study are well established, it is possible that other moderators exist. Future studies can examine possible personality traits, learning orientations, or situational constraint effects as boundary conditions modulating the effects of rationality and morality on ChatGPT-based plagiarism. The effects of social antecedents, such as social support, norms, and group membership or status on ChatGPT-based plagiarism could also be explored. Fourth, researchers

could examine the downstream effects of GPT-based plagiarism, such as written communication and critical thinking skills, as well as the perceived value of higher education. Fifth, researchers could explore educators' perspectives and analyse their grading standards in response to AI-based plagiarism. Namely, educators may be more lenient in their grading if they believe that a written assignment was completed without the assistance of AI tools. Lastly, future research could investigate how the variation in access to educational resources impacts students' likelihood to use AI tools for plagiarism, either intentionally or unintentionally.

Note

1. The research project was approved by the Institutional Review Board committee of the authors' institution. All respondents were provided an informed consent form on the first page of the online survey. They were asked to check a box if they consented to the terms in the form.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix

Perceived punishment severity (Arcy et al. 2009)

If I were caught submitting written assignments using ChatGPT, I think the punishment would be (not severe at all — very severe)

If I were caught submitting written assignments using ChatGPT, I would be severely punished (strongly disagree — strongly agree)

Perceived punishment certainty (Arcy et al. 2009)

If I submitted written assignments using ChatGPT, the probability that I would be caught is (very low – very high)

If I submitted written assignments using ChatGPT, I would probably be caught. (strongly disagree – strongly agree)

Perceived benefits (self-developed)

Using ChatGPT increases my chances of getting a good grade on written assignments.

ChatGPT generates better essays than I can do on my own.

Using ChatGPT helps me finish written assignments more quickly.

Informal sanction (Bossler 2021)

I would feel guilty if I submitted written assignments done by ChatGPT as my own.

I would feel embarrassed if my friends found out that I submitted written assignments done by ChatGPT as my own.

I would feel embarrassed if my parents found out that I submitted written assignments done by ChatGPT as my own.

Moral disengagement towards cheating (Shu, Gino, and Bazerman 2011)

Sometimes getting ahead of the curve is more important than adhering to rules.

Rules should be flexible enough to be adapted to different situations.

Cheating is appropriate behaviour because no one gets hurt.

If others engage in cheating behaviour, then the behaviour is morally permissible.

It is appropriate to seek short-cuts as long as it is not at someone else's expense.

End results are more important than the means by which one pursues those results.

Intention to use ChatGPT (self-developed)

I intend to use ChatGPT for written assignments in the future.

I plan to use ChatGPT frequently for written assignments.

I will try to use ChatGPT for written assignments.

ChatGPT plagiarism threshold (self-developed)

Please choose a number between 0 and 100 for the following question: - It would be acceptable to submit an essay if the ___ % of the content is written by ChatGPT.